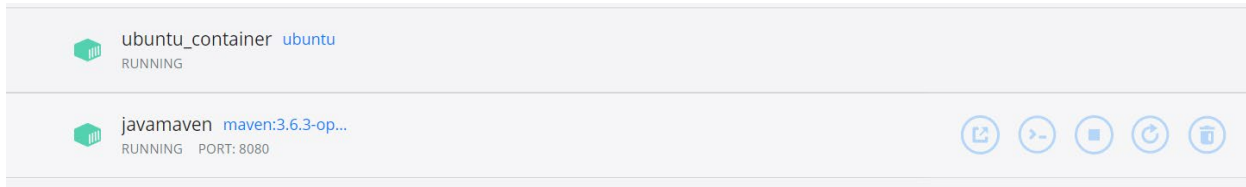


Activity 14.2 Solution File

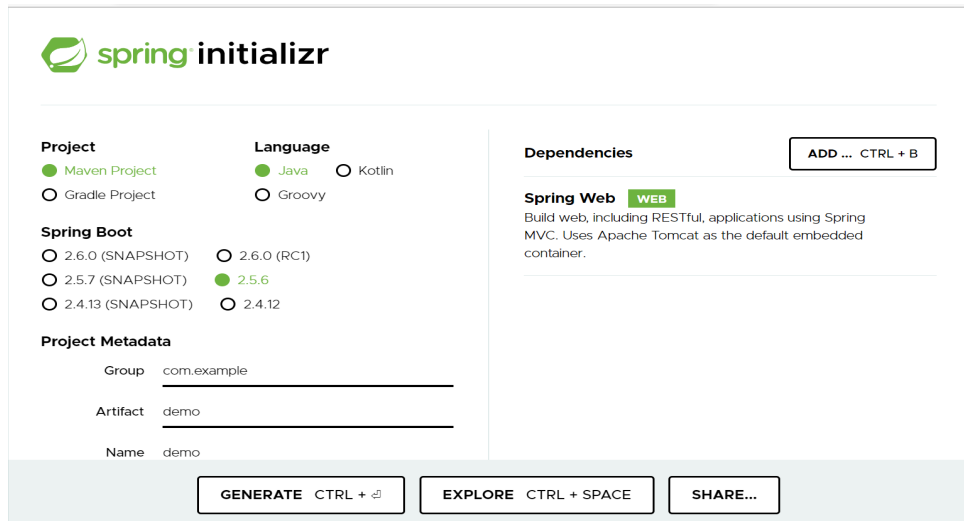
1.



2.

```
docker exec -it 5bd420feefa6c1a9673bd0105cfd5b4a44e6eef31733838d1290f100e13bfbf1 /bin/sh
# apt-get update
Get:1 http://deb.debian.org/debian buster InRelease [122 kB]
Get:2 http://security.debian.org/debian-security buster/updates InRelease [65.4 kB]
Get:3 http://deb.debian.org/debian buster-updates InRelease [51.9 kB]
Get:4 http://security.debian.org/debian-security buster/updates/main amd64 Packages [309 kB]
Get:5 http://deb.debian.org/debian buster/main amd64 Packages [7906 kB]
Get:6 http://deb.debian.org/debian buster-updates/main amd64 Packages [15.2 kB]
Fetched 8469 kB in 2s (5092 kB/s)
Reading package lists... Done
# apt-get install nano
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  spell
The following NEW packages will be installed:
  nano
0 upgraded, 1 newly installed, 0 to remove and 21 not upgraded.
Need to get 544 kB of archives.
After this operation, 2269 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian buster/main amd64 nano amd64 3.2-3 [544 kB]
Fetched 544 kB in 0s (5702 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package nano.
(Reading database ... 12572 files and directories currently installed.)
Preparing to unpack .../archives/nano_3.2-3_amd64.deb ...
Unpacking nano (3.2-3) ...
Setting up nano (3.2-3) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
#
```

3.



The Spring Initializr web form is shown with the following configuration:

- Project:** ☒ Maven Project, ☐ Gradle Project
- Language:** ☒ Java, ☐ Kotlin, ☐ Groovy
- Spring Boot:** ☐ 2.6.0 (SNAPSHOT), ☐ 2.6.0 (RC1), ☐ 2.5.7 (SNAPSHOT), ☒ 2.5.6, ☐ 2.4.13 (SNAPSHOT), ☐ 2.4.12
- Project Metadata:**
 - Group: com.example
 - Artifact: demo
 - Name: demo
- Dependencies:** **Spring Web** (WEB)

Buttons at the bottom: GENERATE (CTRL + G), EXPLORE (CTRL + SPACE), SHARE...

4.

```
C:\tmp\MO-PCDE\demo>dir
Volume in drive C is OS
Volume Serial Number is 92CD-86B9

Directory of C:\tmp\MO-PCDE\demo

11/18/2021  02:16 AM    <DIR>        .
11/18/2021  02:16 AM    <DIR>        ..
11/18/2021  02:16 AM                395  .gitignore
11/18/2021  02:16 AM    <DIR>        .mvn
11/18/2021  02:16 AM                891  HELP.md
11/18/2021  02:16 AM            10,070 mvnw
11/18/2021  02:16 AM            6,608 mvnw.cmd
11/18/2021  02:16 AM            1,226 pom.xml
11/18/2021  02:16 AM    <DIR>        src
               5 File(s)          19,190 bytes
               4 Dir(s)  56,989,184,000 bytes free
```

5.

```
C:\tmp\MO-PCDE\demo>cd ..
C:\tmp\MO-PCDE>dir
Volume in drive C is OS
Volume Serial Number is 92CD-86B9

Directory of C:\tmp\MO-PCDE

11/21/2021  10:20 AM    <DIR>        .
11/21/2021  10:20 AM    <DIR>        ..
11/18/2021  02:16 AM    <DIR>        demo
11/17/2021  09:18 PM            56,863 demo.zip
11/16/2021  12:52 PM            12 foo_container.txt
11/16/2021  12:41 PM            16 foo_local_machine.txt
11/21/2021  10:23 AM    <DIR>        local_activity_14.1
11/16/2021  09:31 AM    <DIR>        minilesson_14.3
               3 File(s)          56,891 bytes
               5 Dir(s)  56,987,037,696 bytes free
```

6.

```
C:\tmp\MO-PCDE>docker cp demo/. javamaven:home/
```

7.

```
# cd home
# ls
# ls
HELP.md  mvnw  mvnw.cmd  pom.xml  src
```

8.

```
# cd src/main/java/com/example/demo
# ls
DemoApplication.java
```

9.

```
# nano DemoApplication.java
# cat DemoApplication.java
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@SpringBootApplication
@RestController
public class DemoApplication {

    public static void main(String[] args)
    {
        SpringApplication.run(DemoApplication.class, args);
    }

    @GetMapping("/hello")
    public String hello(@RequestParam(value = "name", defaultValue="World") String name)
    {
        return String. format("Hello %s!", name);
    }
}
```

10.

```
docker exec -it 5bd420feefa6c1a9673bd0105cfd5b4a44e6eef31733838d1290f100e13bfbf1 /bin/sh
(2.6 MB at 2.1 MB/s)
[INFO] Attaching agents: []

:: Spring Boot :: (v2.5.6)

2021-11-21 16:38:09.347 INFO 376 --- [main] com.example.demo.DemoApplication : Starting DemoApplication using Java 11.0.10 on 5bd420feefa6 with PID 376 (/home/target/classes started by root in /home)
2021-11-21 16:38:09.349 INFO 376 --- [main] com.example.demo.DemoApplication : No active profile set, falling back to default profiles: default
2021-11-21 16:38:09.889 INFO 376 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8080 (http)
2021-11-21 16:38:09.900 INFO 376 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2021-11-21 16:38:09.900 INFO 376 --- [main] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.54]
2021-11-21 16:38:09.940 INFO 376 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2021-11-21 16:38:09.940 INFO 376 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 549 ms
2021-11-21 16:38:10.147 INFO 376 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8080 (http) with context path ''
2021-11-21 16:38:10.155 INFO 376 --- [main] com.example.demo.DemoApplication : Started DemoApplication in 1.096 seconds (JVM running for 1.334)
```

11.

```
< > ↻ 🏠 ⓘ localhost:8080/hello
Hello World!

< > ↻ 🏠 ⓘ localhost:8080/hello?name=John
Hello John!
```